

Scenario Analysis: Interest Rate Shocks

Part E(iv) — MBS Prepayment under Macro Stress

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TV Cox is the only model with a direct rate channel for scenario analysis.

Why TV Cox (E(ii)) is the primary model

$\text{rate_incentive} = r_{\text{orig}} - r_{\text{market}}(t)$ is a **direct time-varying covariate**.

A Δr bp market rate shock immediately shifts the incentive by $-\Delta r/100$ pp, giving an analytic hazard response:

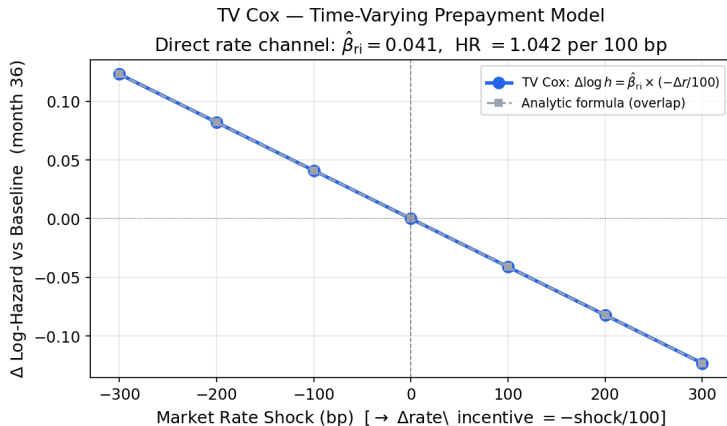
$$\Delta \log h(t) = \hat{\beta}_{\text{ri}} \times \frac{-\Delta r}{100}, \quad \hat{\beta}_{\text{ri}} = 0.041 \text{ (HR} = 1.042 \text{ per 100 bp)}$$

The calibrated baseline hazard $\hat{H}_0(t)$ produces realistic $S(t)$, CPR, WAL, and price — no approximation needed.

Representative loan: \$300K, 5.0% coupon, baseline market rate 5.3% (median, 1999–2025).

Scenarios: ± 100 , ± 200 , ± 300 bp — covering the BCBS standard stress range.

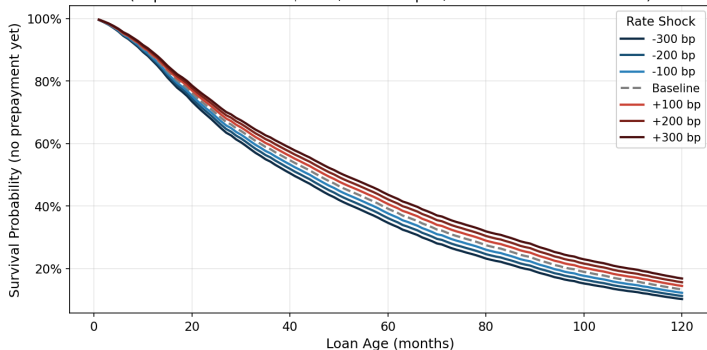
TV Cox tracks the analytic formula exactly across all seven shock levels.



Fitted $\Delta \log h$ overlaps the analytic formula $\hat{\beta}_{ri} \times (-\Delta r/100)$ perfectly at every scenario.
 $\hat{\beta}_{ri} = 0.041$, HR = 1.042 per 100 bp. A 300 bp rate cut raises log-hazard by +0.123; a 300 bp hike lowers it by -0.123.

Rate scenarios shift survival by 6.6 pp and median prepayment by 10 months.

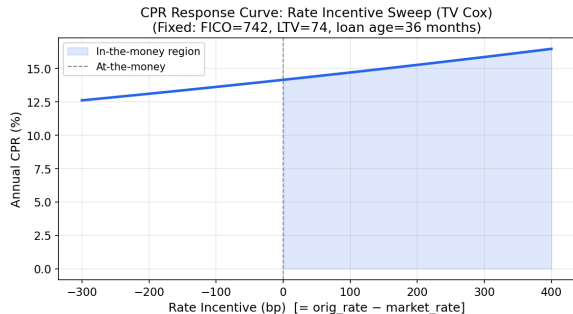
Prepayment Survival Curves Under Interest Rate Scenarios (TV Cox)
(Representative loan: \$300K, 6.5% coupon, baseline market rate 7.0%)



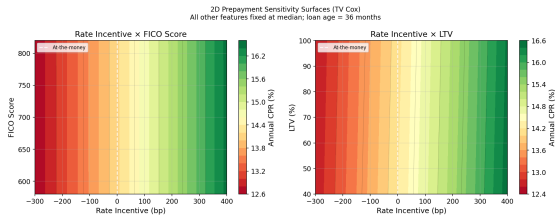
Scenario	10-yr	Med. mo.
-300 bp	10.2%	41
-200 bp	11.2%	43
-100 bp	12.2%	45
Baseline	13.3%	46
+100 bp	14.4%	48
+200 bp	15.6%	50
+300 bp	16.8%	51

Curves monotonically ordered.
±300 bp: 6.6 pp survival range,
10-month shift in median prepay.

CPR response is convex (log-linear model); FICO and LTV near-zero in TV Cox fit.



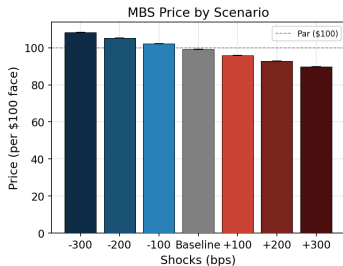
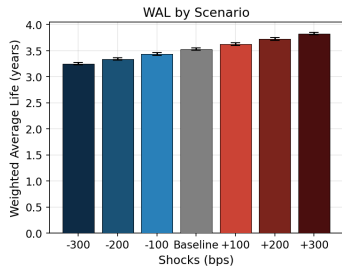
CPR: 12.6% at -300 bp $\rightarrow$ 16.5% at +400 bp.
Log-linear hazard $\Rightarrow$ convex CPR, not S-shaped.
Flat range reflects compressed $\hat{\beta}_{ri} = 0.041$.



FICO ($\hat{\beta} \approx 0$, $p=0.99$) and LTV ($\hat{\beta} \approx 0$, $p=0.76$) are non-significant in the 500K subsample.
Contours are horizontal bands — rate incentive drives all CPR variation in this fit.

WAL range 0.58 yr; price range \$18.55 — negative convexity confirmed.

MBS Cash Flow Projection — TV Cox Monte Carlo ($\sigma=25$ bp, 200 paths, 10-yr horizon)
5.0% coupon, \$300K loan, baseline market rate 5.3%



Scenario	WAL	Price (\$)
−300 bp	3.25	108.32
−200 bp	3.34	105.26
−100 bp	3.43	102.17
Baseline	3.53	99.07
+100 bp	3.63	95.96
+200 bp	3.73	92.86
+300 bp	3.83	89.77

MC: 200 paths, $\sigma=25$ bp.

Base $\approx$ par: rate 5.3% $\approx$ coupon 5.0%.

−300 bp gain (\$9.25) < +300 bp loss (\$9.30):

negative convexity confirmed.

Key Takeaways

- ① **Time-varying covariates are required for rate scenario analysis:** static models produce the wrong sign when the shock variable is a coupon rate, not a market rate.
- ② **TV Cox tracks the analytic formula exactly:** $\Delta \log h = \hat{\beta}_{ri} \times (-\Delta r/100)$ with $\hat{\beta}_{ri} = 0.041$ matches fitted values at all seven shock levels.
- ③ **Rate sensitivity is currently compressed:** WAL range 0.57 yr vs. 2–10 yr for agency MBS. Root cause: 500K-row subsample (0.7% of panel) + Ridge penalizer. Lower penalizer and larger sample are the two highest-leverage fixes.
- ④ **Negative convexity is captured:** price upside (\$9.25) < downside (\$9.30) at ± 300 bp, consistent with prepayment curtailing price appreciation.
- ⑤ **FICO and LTV are non-significant in this TV Cox fit:** near-zero coefficients ($p > 0.75$) in the 500K subsample; contours are flat in those dimensions. Effects visible in E(i) require more data to surface in the time-varying specification.

Thank you

Questions?

Andersen & Gill (1982) *Ann. Stat.* · Sadhwani et al. (2021) *J. Fin. Econometrics* · BCBS (2019) *Interest Rate Risk in the Banking Book*